
SIMBAD PRE-PROCESSING PROGRAM

FOREWORD

THE **SIMBAD** PRE-PROCESSING PROGRAM PROVIDES, FROM THE **PDSTRIP** CALCULATION RESULTS, ALL THE NECESSARY PARAMETERS TO SOLVE THE LINEAR HYDRODYNAMIC MODEL IN THE TIME DOMAIN (**CUMMINS EQUATION**), NAMELY:

- THE “INFINITY” ADDED MASSES
 - THE IMPULSE RESPONSE FUNCTIONS (“IRF”) OF RADIATION DAMPING VALUES
 - THE IMPULSE RESPONSE FUNCTIONS (“IRF”) OF WAVE EXCITATION AND DRIFT FORCES
 - RANDOM WAVE TIME SERIES, ACCORDING TO DIFFERENT OPTIONS:
 - WITH RANDOM PHASE, ON THE BASIS OF H_s , T_p AND DIFFERENT SPECTRUM DENSITY MODELS
 - WITH THE FAST METHOD, ON THE BASIS OF H_s , T_p AND DIFFERENT SPECTRUM DENSITY MODELS
 - WITH RANDOM PHASE, ON THE BASIS OF H_s , T_p AND A SPECIFIC DENSITY SPECTRUM
 - OTHER OPTIONS FOR WAVE TIME SERIES: INPUT AS A SPECIFIC FILE, REGULAR WAVES
 - TIME SERIES OF WAVE EXCITATION FORCES
 - TIME SERIES OF DRIFT FORCES, ON THE BASIS OF THE FOLLOWING OPTIONS:
 - NO DRIFT FORCES
 - MEAN-DRIFT FORCES
 - SLOW-DRIFT LOADS (“MOLIN” VARIANT)
 - TIME SERIES OF WIND VELOCITY ON THE BASIS OF DIFFERENT WIND SPECTRUM DENSITY MODELS
-

NOTICE

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THE INPUT DATA ARE READ FROM THE FILE “PDSTRIP.OUT”.

RESULTS ARE WRITTEN INTO DIFFERENT OUTPUT FILES, THAT ARE FURTHER USED BY THE **SIMBAD** PROGRAM:

- “HOUL_IRR.TXT”, FOR MAIN CHARACTERISTICS OF SHIP (LENGTH, DRAUGHT, POSITION OF CENTRE OF GRAVITY, MASSES AND INERTIA, HYDROSTATICS, “INFINITY” ADDED MASSES...)
- “RAD_IRF.TXT”, FOR THE RADIATION DAMPING IMPULSE RESPONSE FUNCTIONS
- “WAVE_FOR.TXT” AND “DRIFT_FOR.TXT” RESPECTIVELY FOR THE WAVE EXCITATION AND DRIFT FORCES TIME SERIES
- “WIND_VEL.TXT” FOR THE WIND SPEED TIME SERIES

(TWO OTHER FILES “FREQDIR.TXT” AND “RAD_FIT.TXT” ARE ALSO GENERATED, AS INTERMEDIATE DATA FILES)

THESE FILES ARE DESCRIBED IN DETAILS HEREAFTER.

ALL THE INPUTS ARE SPECIFIED IN THE COORDINATE SYSTEMS SUCH AS REQUIRED IN THE **SIMBAD** PROGRAM (SEE DOCUMENTATION OF THIS PROGRAM). THESE COORDINATE SYSTEMS HAVE THE X AXIS POINTING FORWARD, Y TO PORT AND Z UPWARD. THE FIRST COORDINATE SYSTEM (**R1**) HAS ITS ORIGIN LOCATED AT THE INTERSECTION OF REAR SECTION, MIDSHIP PLANE AND WATERLINE PLANE. THE SECOND COORDINATE SYSTEM (**R2**) HAS ITS ORIGIN LOCATED AT THE POSITION OF THE CENTRE OF GRAVITY (WITH ITS X, Y AND Z VALUES SPECIFIED AS PER THE FIRST COORDINATE SYSTEM).

RUNNING THE SIMBAD PRE-PROCESSING PROGRAM: AN EXAMPLE OF RUN IS GIVEN HEREAFTER

Reading the PDSTRIP results in the pdstrip.out file

Reading the PDSTRIP results in the C:\acadata\SIMBAD\SEAWAY_Traitement\Entrees_SIMBAD_PDSTRIP_0.2\pdstrip.out file,

Enter a minimum value for the added masses,
Below this value, the added mass values will not be considered,
ATTENTION!, it will be necessary to check with the graphs,
that one does not lose/gain information with a too high/low value

Enter the minimum value (Default value: 1000 t or tm or tm2):

Enter a minimum value for the radiation dampings,
Below this value, the radiation damping values will not be considered,
ATTENTION!, it will be necessary to check with the graphs,
that one does not lose/gain information with a too high/low value

Enter the minimum value (Default value: 1000 t/s or tm/s or tm2/s):

DEFINITION OF MINIMUM THRESHOLD VALUES FOR ADDED MASSES AND RADIATION DAMPINGS

body1:

- * Setting added mass at infinite frequency to added mass at $\omega = 5.002$
- * Setting added mass at zero frequency to added mass at $\omega = 0.079$
- * Setting real and imaginary excitation components.
- * Setting drift forces.

Do you want plots of Added mass coefficients?: Yes or No (default: No)

Plots of Radiation damping coefficients with smoothing or fitting (default: 1):

- 1: Fitting to an exponential curve,
- 2: Smoothing with hamming method,

Enter the value of frequency limit in rad/s (Default value: 1.75 rad/s):

Do you want plots of Radiation damping coefficients?: Yes or No (default: No)

Replace the Radiation damping values by the fitted ones?: Yes or No (default: No)

Do you want plots of Excitation force real and imag. values?: Yes or No (default: No)

Do you want plots of Drift force?: Yes or No (default: No)

Do you want a Simbad file for irregular wave?: Yes or No (default: Yes)

RECALL OF SOME PARAMETERS OF "PDSTRIP.OUT" FILE

POSSIBILITY OF PLOTS OF DIFFERENT PARAMETERS: ADDED MASSES, RADIATION DAMPING VALUES, EXCITATION AND DRIFT FORCES. AT THE SAME TIME AS THE PLOTTING OF THE PARAMETERS, IT IS PROPOSED A SMOOTHING OF THE VALUES AND POSSIBLY A REPLACEMENT BY THE SMOOTHED VALUES. FOR THE RADIATION DAMPING VALUES, TWO OPTIONS ARE PROPOSED : A SMOOTHING OF VALUES OR A FITTING TO AN EXPONENTIAL CURVE (FROM A FREQUENCY LIMIT : 1.75 RAD/S BY DEFAULT). SEE NOTE ON VALUES REPLACEMENT HEREAFTER.

RUNNING THE SIMBAD PRE-PROCESSING PROGRAM:

NOTE ABOUT THE SMOOTHING OR FITTING OF THE PARAMETERS:

THE SMOOTHING OR FITTING OF THE DIFFERENT PARAMETERS WAS INITIALLY INTENDED TO IMPROVE THE RESOLUTION OF THE LINEAR HYDRODYNAMIC MODEL IN THE TIME DOMAIN.

*HOWEVER, AT THE TIME OF THE COMPARISONS WITH THE RESULTS PUBLISHED IN THE LITERATURE, IT COULD NOT BE SHOWN THAT THESE SMOOTHING OR FITTING COULD IMPROVE THE RESULTS OF THE **SIMBAD** CALCULATIONS. THIS IS UNDOUBTEDLY DUE TO THE FACT THAT FOR THE ADDED MASS, WE USE A VALUE CALLED AT “INFINITY” AND TO SOME EXTENT, FOR THE RADIATION DAMPING VALUES AND THE WAVE EXCITATION AND DRIFT FORCES, WE USE THE IMPULSE RESPONSE FUNCTIONS WHICH WOULD TEND TO SMOOTH THE RESULTS.*

*THE PLOT OF THE VALUES OF THE PARAMETERS IS THEREFORE MAINLY USED TO CHECK THAT THE RESULTS OF THE **PDSTRIP** PROGRAM DO NOT PRESENT A STRONG INCONSISTENCY.*

RUNNING THE SIMBAD PRE-PROCESSING PROGRAM (FOLLOWING):

AN EXAMPLE OF RUN IS GIVEN HEREAFTER

```
Do you want a Simbad file for irregular wave?: Yes or No (default: Yes)

Wave directions:
[ 0. 30. 60. 90. 120. 150. 180. 210. 240. 270. 300. 330.]
Wave direction chosen (default value: 90): 135

Wave frequencies:
[0.079 0.105 0.152 0.2 0.248 0.296 0.345 0.394 0.443 0.493 0.543 0.593
0.644 0.695 0.746 0.798 0.851 0.903 0.955 1.007 1.06 1.112 1.164 1.216
1.267 1.318 1.369 1.42 1.471 1.522 1.572 1.623 1.673 1.724 1.774 1.825
1.875 1.926 1.976 2.026 2.077 2.127 2.177 2.228 2.278 2.329 2.379 2.43
2.48 2.53 2.581 2.631 2.682 2.732 2.783 2.833 2.884 2.934 2.984 3.035
3.085 3.136 3.186 3.236 3.287 3.337 3.388 3.438 3.488 3.539 3.589 3.639
3.69 3.74 3.791 3.841 3.891 3.942 3.992 4.043 4.093 4.144 4.194 4.244
4.295 4.345 4.396 4.446 4.497 4.547 4.598 4.648 4.697 4.748 4.8 4.851
4.898 4.95 5.002]
Wave frequency chosen (default value: 0.5): 0.52

Wave height Hs in m (default value:0.001 m): 1
Do you want a calculation of roll damping coefficients?: Yes or No (default: Yes)

Natural roll period (in s): 11.88
Do you need to change the irregular wave parameters?: Yes or No (default: No)

Interpolation of coefficients in frequency
Interpolation of coefficients in direction

Excitation force IRF for body1:N/A%|
##### |Excitation force IRF for body1:
#|
1: No wave drift forces,
2: Mean wave drift forces,
3: Calculation with Molin formulation,
Default: 3
2
Drift force IRF for body1:N/A%|
##### |Drift force IRF for body1: 98%|#####
Do you want plots of Wave excitation IRF?: Yes or No (default: No)

Do you want plots of drift force IRF?: Yes or No (default: No)

Calculation range for the radiation damping IRF (between 20s and 95s; default value: 20s):
Time step for the radiation damping IRF0.05 to 0.2 s (default value: 0.2s):
Radiation damping IRF for body1:N/A%|##### |Radiation dam
97%|##### |Radiation damping IRF for body1:100%|#####
Do you want plots of Radiation damping IRF?: Yes or No (default: No)

Writing HDF5 data to C:\acadata\SIMBAD\SEAWAY_Traitement\Entrees_SIMBAD_PDSTRIP_0.2\pdstrip.h5
Radiation state space coefficients for body1 were not written.
```

DEFINITION OF DIRECTION, FREQUENCY ($2\pi/T$) AND SIGNIFICANT HEIGHT OF THE WAVE TIME-SERIES

DETERMINATION OF ROLL DAMPING COEFFICIENTS AND NATURAL ROLL PERIOD

INTERPOLATION IN FREQUENCY AND DIRECTION

DETERMINATION OF EXCITATION AND DRIFT FORCES IRFs, AS WELL AS RADIATION DAMPING IRF. FOR THE DRIFT FORCES, A CHOICE IS TO BE DONE, ACCORDING TO 3 OPTIONS.: NO DRIFT FORCES; MEAN WAVE DRIFT FORCES (DRIFT LOADS ARE DERIVED ONLY FROM THE DIAGONAL TERMS OF THE QUADRATIC TRANSFER FUNCTIONS (QTFs)); SLOW DRIFT LOADS, THAT ARE COMPUTED ON THE BASIS OF NEWMAN'S APPROXIMATION AND ACCORDING TO THE MOLIN FORMULATION (SEE : MOLIN B., "SECOND-ORDER HYDRODYNAMICS APPLIED TO MOORED STRUCTURES", BRITISH MARITIME TECHNOLOGY, 1994)

RUNNING THE SIMBAD PRE-PROCESSING PROGRAM (FOLLOWING):

AN EXAMPLE OF RUN IS GIVEN HEREAFTER

Writing HDF5 data to C:\acadata\SIMBAD\SEAWAY_Traitement\Entrees_SIMBAD_PDSTRIP_0.2\pdstrip.h5

Radiation state space coefficients for body1 were not written.

Wave height H_s in m: 1.0 m

Peak period T_p in s: 12.08 s

Wave directions:

[0. 30. 60. 90. 120. 150. 180. 210. 240. 270. 300. 330.]

Wave direction chosen: 135.0 °

Simulation duration in s (default value: 10800):

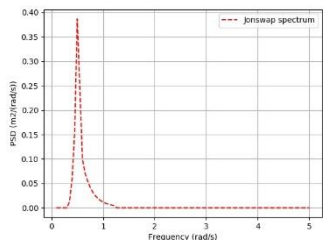
Time step between 0.1 and 1 s (default value: 0.5):

- 1: Jonswap spectrum density model,
 - 2: Bretschneider spectrum density model,
 - 3: Torsethaugen spectrum density model,
 - 4: McCormick spectrum density model,
 - 5: OchiHubble spectrum density model,
 - 6: Wallop spectrum density model,
- Default: 1

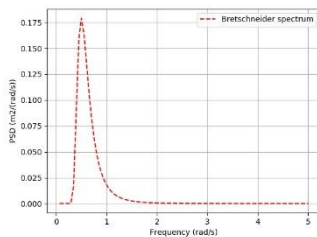
RECALL OF WAVE PARAMETERS PREVIOUSLY CHOSEN

DETERMINATION OF TIME SERIES DURATION AND TIME STEP. A DURATION OF 3 HOURS HAS BEEN TAKEN AS DEFAULT VALUE, AS WELL AS A TIME STEP OF 0.5s. DIFFERENT OPTIONS ARE ALSO POSSIBLE FOR THE WAVE TIME SERIES (SEE HEREAFTER)

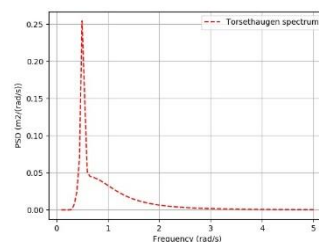
DIFFERENT TYPES OF SPECTRUM DENSITY MODEL ARE AVAILABLE FOR OPTIONS 1 AND 2 OF WAVE TIME SERIES



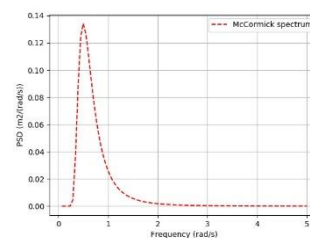
JONSWAP SPECTRUM



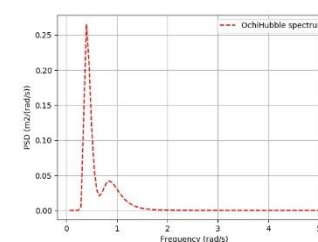
BRETSCHNEIDER SPECTRUM



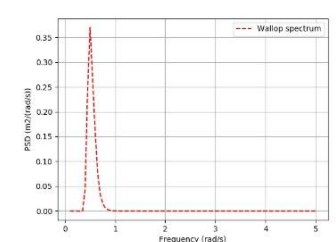
TORSETHAUGEN SPECTRUM



MCCORMICK SPECTRUM



OCHIHUBBLE SPECTRUM



WALLOP SPECTRUM

RUNNING THE SIMBAD PRE-PROCESSING PROGRAM (FOLLOWING):

AN EXAMPLE OF RUN IS GIVEN HEREAFTER

```
1: Wave time series with random phase,  
2: Wave time series with the Fast method,  
3: Wave time series from a specific file,  
4: Wave time series from a specific density spectrum,  
5: Regular waves (Amplitude=0.5*Hs),  
Default: 2
```

DIFFERENT OPTIONS ARE POSSIBLE FOR THE WAVE TIME SERIES: OPTION 1 COMPUTES A WAVE ELEVATION WITH A RANDOM PHASE ; OPTION 2 CALCULATES A GAUSSIAN-DISTRIBUTED WAVE ELEVATION WITH A UNIFORMLY DISTRIBUTED RANDOM PHASE AND ALSO A NORMALLY DISTRIBUTED AMPLITUDE ; OPTION 3 CONSIDERS A WAVE TIME SERIES AS INPUT; OPTION 4 IS SIMILAR TO OPTION 1 BUT FOR A DENSITY SPECTRUM GIVEN AS INPUT; OPTION 5 IS FOR REGULAR WAVES.

THE PREVIOUS CHOICE OF A SPECTRUM DENSITY MODEL CONCERNS ONLY THE OPTIONS 1 AND 2

```
Wave height Hs in m: 1.000 m  
Wave height Hmax in m: 1.879 m  
Mean wave period T1 in s: 9.679 s  
Peak wave period Tp in s: 12.800 s  
Do you want plots of Wave excitation forces?: Yes or No (default: No)
```

CHARACTERISTICS OF CALCULATED WAVE TIME SERIES

```
Do you want plots of Drift excitation forces?: Yes or No (default: No)
```

PLOTS OF WAVE EXCITATION AND DRIFT FORCES

RUNNING THE SIMBAD PRE-PROCESSING PROGRAM:

NOTE ON THE CHOICE OF OPTIONS FOR THE WAVE TIME –SERIES:

THE VAST MAJORITY OF THE COMPARISONS MADE WITH THE RESULTS PUBLISHED IN THE LITERATURE WERE MADE BY TAKING INTO ACCOUNT OPTION 2 (FAST METHOD) FOR THE CALCULATION OF THE TIME SERIES OF WAVES AND FORCES. IT IS THEREFORE ADVISABLE TO TAKE THIS OPTION INTO ACCOUNT IN THE PRESENCE OF WAVES, WHEN THE JONSWAP SPECTRUM IS TAKEN INTO ACCOUNT.

THE STRIP METHOD USED FOR THE HYDRODYNAMIC CALCULATIONS CAN ONLY PROVIDE APPROXIMATE VALUES FOR THE DRIFT FORCES. OPTION 3 OF CALCULATING THE DRIFT FORCES (MOLIN FORMULATION) SOMETIMES GIVES QUITE STRONG RESULTS, ESPECIALLY IN COMBINATION WITH GUSTY WIND. FOR NOW, IT IS RECOMMENDED TO CHOOSE OPTION 3 IF THE WIND IS NOT TO BE CONSIDERED AND OPTION 2 OTHERWISE.

NOTE ON THE CHOICE OF TIME –SERIES DURATION AND TIME STEP:

THE DURATION SETS THE LENGTH OF THE INCIDENT WAVE KINEMATICS TIME SERIES, BUT IT ALSO DETERMINES THE FREQUENCY STEP USED IN THE INVERSE FOURIER TRANSFORM, FROM WHICH THE TIME SERIES ARE DERIVED (WITH OPTION 2). IF THE TOTAL SIMULATION TIME IS MORE THAN THE DURATION OF THE WAVE TIME SERIES, THERE WILL BE A CYCLIC REPEATABILITY OF THE WAVE TRAIN. THE TIME STEP FOR THE WAVE TIME SERIES DETERMINES ALSO THE MAXIMUM FREQUENCY IN THE INVERSE FOURIER TRANSFORM. WHEN MODELLING IRREGULAR SEA STATES, IT IS RECOMMENDED THAT THE DURATION BE SET TO AT LEAST 3 HOURS (DNV, BV) AND THAT THE TIME STEP BE A VALUE IN THE RANGE BETWEEN 0.1 AND 1 s. THIS TIME STEP IS SPECIFIED INDEPENDENTLY FROM THE TIME STEP OF SIMBAD PROGRAM (WAVE FORCES ARE INTERPOLATED IN TIME AS NECESSARY). IN SIMBAD PROGRAM, THE MAXIMUM NUMBER OF TIME STEPS IS 28800, I.E.: 4 HOURS AT 0.5s TIME STEP OR 2 HOURS AT 0.25s TIME STEP.

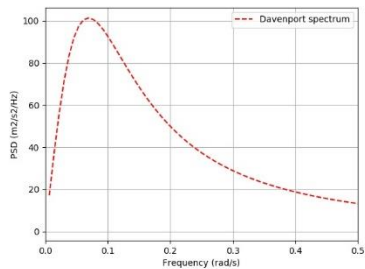
RUNNING THE SIMBAD PRE-PROCESSING PROGRAM (FOLLOWING):

AN EXAMPLE OF RUN IS GIVEN HEREAFTER

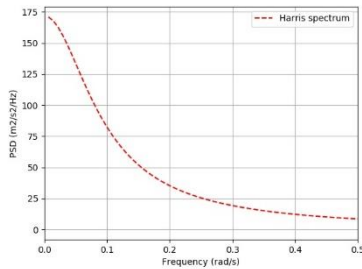
```
Ten min. mean wind velocity at 10m height in m/s(default value: 0.0): 17
Wind direction at 10m height in °(default value: 0.0): 210
Simulation duration in s (default value: 10800):
Time step between 0.5 and 2 s (default value: 1):
0: No wind,
1: Davenport spectral density model,
2: Harris spectral density model,
3: Kaimal spectral density model,
4: API spectral density model,
5: NPD spectral density model,
6: Ochi-Shin spectral density model,
Default: 0
1
Press the ENTER key to continue...
>>> |
```

*DEFINITION OF 10' MEAN WIND SPEED,
WIND DIRECTION, DURATION OF TIME SERIES
AND TIME STEP*

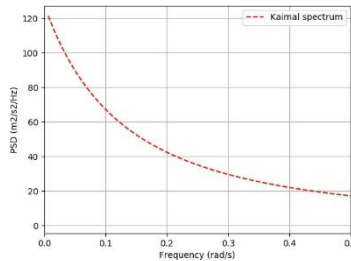
*DIFFERENT TYPES OF SPECTRUM DENSITY MODEL ARE AVAILABLE FOR THE
WIND VELOCITY TIME SERIES*



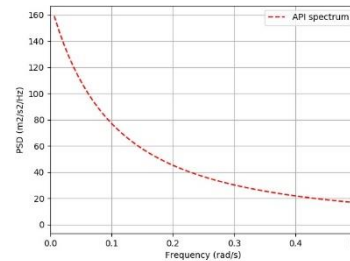
DAVENPORT SPECTRUM



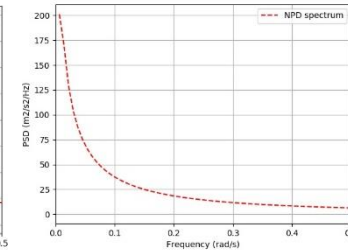
HARRIS SPECTRUM



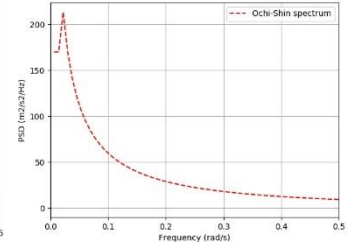
KAIMAL SPECTRUM



API SPECTRUM



*NPD SPECTRUM
(NOT RECOMMENDED)*



OCHI-SHIN SPECTRUM

RUNNING THE SIMBAD PRE-PROCESSING PROGRAM:
RESULT FILE “HOUL_IRR.TXT”, FOR MAIN CHARACTERISTICS OF SHIP

SHIP'S LPP/NUMBER OF BODIES/XG, YG, ZG OF CENTRE OF GRAVITY IN COORDINATE SYSTEM R1					
WAVE Tp/DIRECTION/Hs					
SHIP'S DRAFT/ROLL DAMPING COEFFICIENTS/ FENDER FRICTION COEFFICIENT					
387.50	1	180.80	0.00	11.20	
8.00		90.00	0.01		
13.00	8.14998823e+08	1.00974232e+10		0.18	
2.03388e+08	0.00000e+00	0.00000e+00	0.00000e+00	2.44827e+02	0.00000e+00
0.00000e+00	2.03388e+08	0.00000e+00	-2.44827e+02	0.00000e+00	0.00000e+00
0.00000e+00	0.00000e+00	2.03388e+08	0.00000e+00	0.00000e+00	0.00000e+00
0.00000e+00	-2.44827e+02	0.00000e+00	8.96941e+10	0.00000e+00	0.00000e+00
2.44827e+02	0.00000e+00	0.00000e+00	0.00000e+00	1.83558e+12	0.00000e+00
0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	1.83558e+12
0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00
0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00
0.00000e+00	0.00000e+00	1.94804e+08	0.00000e+00	1.68186e+09	0.00000e+00
0.00000e+00	0.00000e+00	0.00000e+00	1.72186e+10	0.00000e+00	0.00000e+00
0.00000e+00	0.00000e+00	1.68186e+09	0.00000e+00	1.79946e+12	0.00000e+00
0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00
5.114163e+06	0.000000e+00	0.000000e+00	0.000000e+00	-5.739518e+07	0.000000e+00
0.000000e+00	5.582149e+07	0.000000e+00	-5.213962e+08	0.000000e+00	-7.276625e+07
0.000000e+00	0.000000e+00	1.257570e+09	0.000000e+00	1.207145e+10	0.000000e+00
0.000000e+00	-5.269824e+08	0.000000e+00	4.630606e+10	0.000000e+00	3.014874e+10
-5.739518e+07	0.000000e+00	1.207145e+10	0.000000e+00	6.862909e+12	0.000000e+00
0.000000e+00	-7.276641e+07	0.000000e+00	3.000356e+10	0.000000e+00	6.816518e+11

MASSSES, INERTIA

HYDROSTATICS

“INFINITY” ADDED MASSES

RUNNING THE SIMBAD PRE-PROCESSING PROGRAM: RESULT FILE “RAD_IRF.TXT”, FOR THE RADIATION DAMPING IMPULSE RESPONSE FUNCTIONS

```
101  
15  
0 0  
0.000000e+00 1.731438e+07  
2.000000e-01 1.701936e+07  
4.000000e-01 1.615220e+07  
6.000000e-01 1.476519e+07  
8.000000e-01 1.294108e+07  
1.000000e+00 1.078683e+07  
1.200000e+00 8.425630e+06  
1.400000e+00 5.987904e+06  
1.600000e+00 3.601918e+06  
1.800000e+00 1.384903e+06  
2.000000e+00 -5.646791e+05  
2.200000e+00 -2.173095e+06  
2.400000e+00 -3.395116e+06  
2.600000e+00 -4.215364e+06  
2.800000e+00 -4.647225e+06  
3.000000e+00 -4.729540e+06  
3.200000e+00 -4.521436e+06  
3.400000e+00 -4.095865e+06  
3.600000e+00 -3.532449e+06  
3.800000e+00 -2.910308e+06  
4.000000e+00 -2.301490e+06  
4.200000e+00 -1.765564e+06  
4.400000e+00 -1.345755e+06  
4.600000e+00 -1.066895e+06  
4.800000e+00 -9.352444e+05  
5.000000e+00 -9.400533e+05  
5.200000e+00 -1.056614e+06  
5.400000e+00 -1.250385e+06  
5.600000e+00 -1.481721e+06  
5.800000e+00 -1.710692e+06  
6.000000e+00 -1.901515e+06  
6.200000e+00 -2.026166e+06  
6.400000e+00 -2.066887e+06  
6.600000e+00 -2.017381e+06  
6.800000e+00 -1.882683e+06  
7.000000e+00 -1.677802e+06
```

NUMBER OF TIME STEPS FOR EACH IRF

TOTAL NUMBER OF IRFs

POSITION OF IRF IN MATRIX 6X6 (FROM 0 TO 5)

VALUES OF IRF AT EACH TIME STEP

RUNNING THE SIMBAD PRE-PROCESSING PROGRAM: RESULT FILE “WAVE_FOR.TXT” FOR THE WAVE EXCITATION FORCES TIME SERIES

```
21600
0
0.000000e+00 -1.201503e+05
5.000000e-01 -6.441925e+05
1.000000e+00 -1.141075e+06
1.500000e+00 -1.603663e+06
2.000000e+00 -2.033403e+06
2.500000e+00 -2.432094e+06
3.000000e+00 -2.793961e+06
3.500000e+00 -3.101305e+06
4.000000e+00 -3.325827e+06
4.500000e+00 -3.434935e+06
5.000000e+00 -3.400360e+06
5.500000e+00 -3.206678e+06
6.000000e+00 -2.856275e+06
6.500000e+00 -2.369747e+06
7.000000e+00 -1.781830e+06
7.500000e+00 -1.134504e+06
8.000000e+00 -4.694250e+05
8.500000e+00 1.784620e+05
9.000000e+00 7.846083e+05
9.500000e+00 1.335203e+06
1.000000e+01 1.824195e+06
1.050000e+01 2.248018e+06
1.100000e+01 2.599791e+06
1.150000e+01 2.865455e+06
1.200000e+01 3.023426e+06
1.250000e+01 3.048597e+06
1.300000e+01 2.919710e+06
1.350000e+01 2.627226e+06
1.400000e+01 2.178591e+06
1.450000e+01 1.598421e+06
1.500000e+01 9.235067e+05
1.550000e+01 1.946726e+05
1.600000e+01 -5.508071e+05
1.650000e+01 -1.282918e+06
1.700000e+01 -1.978010e+06
1.750000e+01 -2.614999e+06
1.800000e+01 -3.170842e+06
```

NUMBER OF TIME STEPS FOR EACH TIME SERIES

DEGREE OF FREEDOM OF WAVE FORCE (FROM 0 TO 5)

VALUES OF WAVE EXCITATION FORCE AT EACH TIME STEP

RUNNING THE SIMBAD PRE-PROCESSING PROGRAM: RESULT FILE “DRIFT_FOR.TXT” FOR THE DRIFT FORCES TIME SERIES

```
21600
0
0.000000e+00 -4.878323e+03
5.000000e-01 -8.189604e+03
1.000000e+00 -9.793693e+03
1.500000e+00 -9.782900e+03
2.000000e+00 -9.364010e+03
2.500000e+00 -8.490547e+03
3.000000e+00 -7.396896e+03
3.500000e+00 -6.138132e+03
4.000000e+00 -5.003919e+03
4.500000e+00 -3.768376e+03
5.000000e+00 -2.616943e+03
5.500000e+00 -1.623492e+03
6.000000e+00 -9.502779e+02
6.500000e+00 -6.361537e+02
7.000000e+00 -5.442384e+02
7.500000e+00 -6.215640e+02
8.000000e+00 -7.475032e+02
8.500000e+00 -9.058108e+02
9.000000e+00 -1.032087e+03
9.500000e+00 -1.079840e+03
1.000000e+01 -9.695234e+02
1.050000e+01 -7.771968e+02
1.100000e+01 -4.599808e+02
1.150000e+01 -2.253426e+02
1.200000e+01 -1.531199e+02
1.250000e+01 -2.746793e+02
1.300000e+01 -5.457013e+02
1.350000e+01 -8.858615e+02
1.400000e+01 -1.213075e+03
1.450000e+01 -1.464472e+03
1.500000e+01 -1.641396e+03
1.550000e+01 -1.711350e+03
1.600000e+01 -1.759613e+03
1.650000e+01 -1.814918e+03
1.700000e+01 -1.893054e+03
1.750000e+01 -2.057368e+03
1.800000e+01 -2.301593e+03
-
-
```

NUMBER OF TIME STEPS FOR EACH TIME SERIES

DEGREE OF FREEDOM OF DRIFT FORCE (FROM 0 TO 5)

VALUES OF DRIFT FORCE AT EACH TIME STEP

RUNNING THE SIMBAD PRE-PROCESSING PROGRAM :
RESULT FILE “WIND_VEL.TXT” FOR THE WIND VELOCITY TIME SERIES

210.0000	RELATIVE WIND DIRECTION
0.00	10.22
2.00	13.02
4.00	16.32
6.00	18.22
8.00	20.16
10.00	19.90
12.00	19.19
14.00	23.30
16.00	18.96
18.00	19.95
20.00	22.05
22.00	16.86
24.00	18.02
26.00	13.48
28.00	15.77
30.00	20.65
32.00	18.27
34.00	18.07
36.00	19.52
38.00	16.74
40.00	18.35
42.00	16.90
44.00	15.13
46.00	20.91
48.00	17.20
50.00	20.08
52.00	20.88
54.00	17.19
56.00	17.49
58.00	16.30
60.00	19.09
62.00	22.00
64.00	17.98
66.00	13.65
-	
-	
-	

WIND SPEED TIME SERIES (TIME IN S VS. WIND SPEED IN M/S). IT IS MANDATORY TO END WITH A NEGATIVE TIME VALUE. THE NUMBER OF TIME STEPS IS LIMITED TO 28800. IF THE SIMULATION TIME IS GREATER THAN THE DURATION OF THE TIME SERIES, THE SERIES IS REPEATED CYCLICALLY. THE DURATION OF TIME SERIES SHALL BE LESS THAN (OR EQUAL TO) THE TOTAL SIMULATION TIME.